

Mecha-medical!

Playing games like *Command & Conquer* and *Age of Empires* would never have been possible without those mobile repair stations at the battlefield.



An artist's impression of the Mobile Asset Repair Systems of the future

Plasma Rifle

A constant source of enjoyment in any game—from the BFG10k gun in *Doom II* to those used in *Unreal Tournament*, and *Quake* (sorry, we couldn't resist!), plasma rifles are the absolute future in weapons inventory, and are normally accompanied by bulky-looking guns spewing out obscene amounts of green or blue pulses of energy in a game.

Plasma-based weaponry is theoretically any group of weapons designed to use highly charged, energy-rich ionised gas or "plasma". The idea behind the plasma rifle is that the energy is aimed not directly at the target but a little ahead of it. Since the beams would travel at the speed of light, they would, in theory, be able to intercept any target flying at sub-sonic or even hyper-sonic speed. A cloud of highly-charged ionised air forms at the target focus area. When it enters the cloud, any object—such as a missile or airplane—is deflected from its current course and breaks



The Plasma Gun in all its glory!

What fun is it if you have to wait a few hours till you re-spawn your armies for your next conquest? We hate trainers and cheat-codes...

As if taking a cue from the games themselves, scientists at the United States Air Force are building a mobile facility that could be located just near the battlefield—where soldiers can be patched up, or parts repaired or manufactured in real-time. Medkits, anyone?

The system has been christened Mobile Asset Repair System (MARS). This facility can be based on land—or in the middle of an ocean. The facility will include a set of Flexible Manufacturing Systems (FMS) and robotic arms that can be linked electronically to commercial suppliers of specific military equipment. A wide array of parts can thus be manufactured. The manufacturers feed in the specifications of the required part—remotely—to the FMS, which then produces it. The manufacturing materials necessary for MARS to operate are proposed to be obtained from the countries in which the system will eventually be deployed.



An artist's impression of the Shiva Star System

down due to the large air pressure differences between the outside atmosphere and the interior of the flying body. So much for the *Star Trek* nay-sayers!

In real life, however, the plasma gun has only come as close as the Shiva Star, an ongoing project for a high-powered plasma beam, with plans of it being used as a tactical weapon in the Strategic Defence Initiative used

to protect the USA from nuclear ballistic missiles. It has been in research since the early 1990s at Kirtland Air Force Base in the US. The Shiva Star is a system that can presumably be mounted on a satellite or spacecraft in order to shoot down incoming missiles with projectiles of plasma travelling at very high velocities, thereby damaging the missile's guidance sensors. Work on this has languished, though: costs are running high.

With all these technologies in place, we can imagine an army recruiter in the future asking what games you've played. But then again, who needs humans if researchers take a cue from the *Worms* series and clone the ultimate Ninja Sheep, capable of throwing explosive Ninja stars across vast distances! Nukes suddenly seem so passé...

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